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Project Name: Jet Cultivation

Experiment 1: Study Of Azure

Aim:

To study Microsoft Azure cloud platform, its benefits, services provided, and compare IaaS, PaaS, and SaaS with examples.

Introduction:

Microsoft Azure is a cloud computing platform provided by Microsoft. It offers a wide range of services such as computing, storage, networking, and analytics through the internet. It helps developers and organizations build, deploy, and manage applications efficiently.

Benefits of Azure:

- Scalability: Easily increase or decrease resources based on demand
- Cost-effective: Pay only for what you use
- Security: Advanced security features and compliance support
- Flexibility: Supports multiple programming languages and frameworks
- Global Availability: Data centers available worldwide

Services Provided by Azure:

Azure provides many services including:

- Virtual Machines (VMs)
- App Services
- Azure Storage
- Azure SQL Database
- Networking Services
- AI and Machine Learning Services

Cloud Service Models:

1. IaaS (Infrastructure as a Service)

- Provides virtualized computing resources over the internet
- Users manage OS, applications, and data

- Example: Azure Virtual Machines

2. PaaS (Platform as a Service)

- Provides platform to develop, run, and manage applications
- No need to manage infrastructure
- Example: Azure App Service

3. SaaS (Software as a Service)

- Provides software applications over the internet
- No need to install or maintain software
- Example: Microsoft 365

Comparison between AWS, Azure and GCP

Feature	AWS	Azure	GCP
Launch Year	2006	2010	2011
Provider	Amazon	Microsoft	Google
Market Share	Highest	Second	Third
Compute	EC2	Virtual Machines	Compute Engine
Storage	S3	Blob Storage	Cloud Storage
Database	RDS, DynamoDB	SQL DB, Cosmos DB	Cloud SQL, Firestore
Pricing	Complex	Flexible	Simple
Best For	Startups	Enterprises	AI/ML
AI/ML	SageMaker	Azure ML	Vertex AI
Integration	Broad	Microsoft tools	Google services
Ease of Use	Moderate	Easy	Simple
Hybrid Cloud	Limited	Strong	Growing
Global Reach	Very Large	Very Large	Good
Security	Strong	Very Strong	Strong

Conclusion

Microsoft Azure provides flexible and scalable cloud solutions. Depending on user needs, IaaS, PaaS, and SaaS offer different levels of control and management, making Azure suitable for various applications and businesses.